Muhammed Yousuf S

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Education

2023-Present	Ph.D. in Environmental Engineering Civil and Environmental Engineering Indian Institute of Technology Delhi	CGPA: 9.25/10
2020-22	Master of Technology Civil Engineering APJ Abdul Kalam Technological University	CGPA: 9.49/10
2015-19	Bachelor of Technology Civil Engineering APJ Abdul Kalam Technological University	CGPA: 7.53/10

Journal Publications

- **Yousuf, S. M.**, Karigowda, Habib, G., Warsi, A. B., Imran, M., & Nabi, M. un. (2026). Chemical investigation of polycyclic aromatic hydrocarbon sources and associated health risks in PM2.5 from Eastern India. *Scientific Reports*. https://doi.org/10.1038/s41598-026-41899-0
- Shibu, K., Drisiya, J., & **Yousuf, S. M.** (2025). Spatiotemporal Modelling Approach for Nutrient Export in Sasthamkotta Freshwater Wetland Watershed. Journal of the Indian Society of Remote Sensing, 53(1), 225–239. https://doi.org/10.1007/s12524-024-01978-z
- Shibu, K., Ayoob, S., & **Yousuf, S. M.** (2023). Determination of land use–land cover changes in the adjoining panchayats of Sasthamkotta freshwater wetland, a Ramsar site in South India. International Journal of Lakes and Rivers, 16(1), 17–33.
- Deepak, K. R., Sabu, P., **Yousuf, S. M.,** & Thottumkal, C. G. (2022). Urban flood susceptibility map of Chennai using GIS and random forest method. International Research Journal of Engineering and Technology, 9(10).

Technical Skills

Instrumentation

- Thermal Desorption Gas Chromatography-Mass Spectrometry (TD-GC/MS)
- Organic Carbon/Elemental Carbon Analyzer (OC/EC Analyzer)
- Ion Chromatography (IC)
- Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES)
- Aethalometer
- Nephelometer
- Scanning Mobility Particle Sizer (SMPS)
- Monitor for AeRosols and Gases in Ambient Air (MARGA)
- Ambient Metals Monitor (Xact)
- Micro-Orifice Uniform Deposit Impactor (MOUDI)

Software and Analytical Tools

- ArcGIS
- QGIS
- MS-Office
- Python
- Origin

Research Presentations

Yousuf, S. M., Habib, G. Seasonal Variation, Ring Distribution, and Source Characteristics of $PM_{2.5}$-bound PAHs in Eastern India. Platform Presentation at Asian Aerosol Conference 1-4 December 2025

Workshops and Trainings

- **Advanced Aerosol Science Intensive Course** (View Certificate)
  Centre for Atmospheric Sciences (CAS), IIT Delhi and Institute for Atmospheric & Earth System Research (INAR), University of Helsinki, 19th – 29th November 2025
- **Minds-on Air quality, humaN healTH & climAte Nexus** (View Certificate)
  Central University of Jammu & Sindhu Central University, 26th February – 1st March 2024
- **Aerosol Science Intensive Course** (View Certificate)
  Centre for Atmospheric Sciences (CAS), IIT Delhi and Institute for Atmospheric & Earth System Research (INAR), University of Helsinki, 5th - 15th February 2024
- **Workshop on source apportionment** (View Certificate)
  Paul Scherrer Institute (PSI) and the partners of the Clean Air Project India – Indian Institute of Technology Bombay, 15th & 16th December 2023.
- **Workshop on WebGIS Enabled Spatial Information System** (View Certificate)
  TPLC, Govt. Engineering College Barton Hill Trivandrum - Directorate of Technical Education, Government of Kerala, 8th – 18th February 2021.

Awards and Achievements

- **Graduate Aptitude Test in Engineering 2019**
  Civil Engineering (Score: 321)
- **Civil Prize 2019**
  National Council for Technology and Training
- **Graduate Aptitude Test in Engineering 2022**
  Geomatics (Score: 450)

Research Experience

2023 - Pres	Investigating the Contribution of Organics in Refining Source Apportionment (PhD Thesis) Advisor: Prof. Gazala Habib	• Investigated the chemical composition of ambient particulate matter through integrated analysis of organic, ionic, elemental, and carbonaceous (OC/EC) species. • Developed TD–GC–MS methods for quantification of 16 EPA PAHs, alkanes, and phthalates. • Applied Positive Matrix Factorization (PMF) for source apportionment and performed data interpretation.
2025	Influence of Black Carbon Mixing State on Mass Absorption Cross-Section in the Indo-Gangetic Plain (Winter Campaign)	• Investigated the physical and optical characteristics of ambient aerosols using SMPS, nephelometer, and black carbon measurements.

- Examined carbonaceous aerosol composition through integration of offline filter sampling and OC/EC analysis with real-time observations.
- Analyzed PTAAM datasets alongside multi-instrument measurements for comprehensive characterization of particulate matter.
- Performed systematic data processing, quality control, and data interpretation to understand aerosol variability and atmospheric processes.

2020-2022 Modelling of Water Quality of Sasthamkotta Lake Using GIS and Remote Sensing (Mtech Thesis)
Advisor: Prof. Shibu Krishnan

- Assessed spatial and temporal variations in water quality of Sasthamkotta Lake using geospatial techniques.
- Applied GIS and remote sensing tools for water quality modelling and mapping.
- Integrated satellite data and field-based observations for environmental analysis.
- Developed thematic maps to identify pollution patterns and critical zones.
- Performed data processing, analysis, and interpretation for environmental monitoring.

Positions of Responsibility

2025 **Teaching Assistantship (IIT Delhi)**

- Advanced Air Pollution Laboratory (CVP 820, Graduate course)
- Contributed to course content development and lecture preparation
- Assisted in class delivery and student engagement
- Supported laboratory experiments and practical sessions
- Assisted in preparing and grading assignments and examinations

2025 **Teaching Assistantship (IIT Delhi)**

- Air Pollution and Control (CVL 7705, Graduate course)
- Assisted in course delivery and classroom instruction
- Conducted lecture sessions and facilitated student discussions
- Prepared and graded assignments and examinations
- Coordinated and organized a student field visit, managing logistics and participant arrangements.

2024	**Teaching Assistantship (IIT Delhi)**	• Advanced Air Pollution Laboratory (CVP 820, Graduate course) • Contributed to course content development and lecture preparation • Assisted in class delivery and student engagement • Supported laboratory experiments and practical sessions • Assisted in preparing and grading assignments and examinations
2024	**Teaching Assistantship (IIT Delhi)**	• Environmental Science (CVL 100, Undergraduate Course) • Assisted in grading exams/assignments and contributed to preparation of question papers.